MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

Guidelines for Treatment of Prosthetic Valve Infective Endocarditis in Patients on Pediatric Services

(Infectious Diseases consultation is STRONGLY recommended)

Empiric therapy (prior to confirmation of pathogen):

- *Early (≤ 1 year since valve replacement):*
  - **Vancomycin** IV[3] + **Cefepime** 50 mg/kg/dose IV q8h[1] (max: 2 g/dose) + **Rifampin** 7 mg/kg/dose PO/IV q8h (max: 300 mg/dose) + **Gentamicin** IV[4]
- *Late (> 1 year since valve replacement):*
  - **Vancomycin** IV[3] + **Ceftriaxone** 100 mg/kg/dose IV q24h (max: 2 g/dose) ± **Rifampin** 7 mg/kg/dose PO/IV q8h (max: 300 mg) ± **Gentamicin** IV[4]

Gentamicin goals:

- Gentamicin is used for gram positive synergy
- Traditional gentamicin synergy dosing (1 mg/kg/dose IV q8h) is preferred in pediatric patients
- Gentamicin peak goal 3-5 mg/L and trough goal < 1 mg/L

Definitive Therapy (Once Pathogen is Identified)				
Pathogens		**Antibiotic Therapy (renal dose adjustments may be necessary)[1]**	**Duration of Therapy**	**Comments**
Viridans group streptococci **OR** ***Streptococcus gallolyticus (bovis)***	Penicillin MIC ≤ 0.12 mg/L	Preferred: **Penicillin G** 50,000 units/kg/dose IV q4h[1,2] (max: 4 million units/dose) ± **Gentamicin** IV[4] for first 2 weeks OR **Ceftriaxone** 100 mg/kg/dose IV q24h (max: 2 g/dose) ± **Gentamicin** IV[4] for first 2 weeks Alternative for Severe PCN Allergy: **Vancomycin** IV[3]	6 weeks	
	Penicillin MIC > 0.12	Preferred: **Penicillin G** 50,000 units/kg/dose IV q4h[1,2] (max: 4 million units/dose) + **Gentamicin** IV[4] OR **Ceftriaxone** 100 mg/kg/dose IV q24h (max: 2 g/dose) + **Gentamicin** IV[4] Alternative for Severe PCN Allergy: **Vancomycin** IV[3]	6 weeks	
Enterococci		See native valve guideline for antimicrobial selection	≥ 6 weeks	• For Enterococci strains resistant to vancomycin, aminoglycosides, and penicillin, > 6 weeks of therapy is recommended (for all other Enterococci strains 6 weeks of therapy should be used).
Staphylococci (MSSA, MSSE)		Preferred: **Nafcillin** 33 mg/kg/dose IV q4h[2] (max: 2 g/dose) + **Rifampin** 7 mg/kg/dose PO q8h (max : 300 mg/dose) + **Gentamicin** IV[4] for first 2 weeks	≥ 6 weeks	• Cefazolin may be used instead of nafcillin in patients with penicillin allergy (non-anaphylaxis) if CNS disease is not present.

Definitive Therapy (Once Pathogen is Identified)			
Pathogens	**Antibiotic Therapy (renal dose adjustments may be necessary)[1]**	**Duration of Therapy**	**Comments**
Staphylococci (MRSA, MRSE)	**Preferred:** **Vancomycin** IV[3] + **Rifampin** 7 mg/kg/dose PO q8h (max : 300 mg/dose) + **Gentamicin** IV[4] for first 2 weeks Alternative for Vancomycin Allergy or Failure: **Daptomycin** 10 mg/kg IV q24h[1] + **Rifampin** 7 mg/kg/dose PO q8h (max: 300 mg/dose) + **Gentamicin** IV[4] for first 2 weeks	≥ 6 weeks	• Follow baseline and weekly CK with daptomycin therapy
HACEK Group	See native valve guideline for antimicrobial selection	6 weeks	
***Candida* spp.**	See native valve guideline for antimicrobial selection	≥ 6 weeks	
Culture negative, early (≤1 year since valve replacement)	**Vancomycin** IV[3] + **Cefepime** 50 mg/kg/dose IV q8h[1] (max: 2 g/dose) + **Rifampin** 7 mg/kg/dose PO q8h (max: 300 mg/dose) + **Gentamicin** IV[4] for first 2 weeks	6 weeks	• Receipt of antibiotics prior to obtaining cultures is the most common cause of culture negative IE. There are many infectious and non-infectious causes. An evaluation of epidemiological factors, history of prior cardiovascular infections, exposure to antimicrobials, clinical course, severity, and extracardiac sites of infection should be performed to help guide diagnosis and treatment.
Culture negative, late (>1 year since valve replacement)	**Vancomycin** IV[3] + **Ceftriaxone** 100 mg/kg/dose IV q24h (max 2 g) ± **Rifampin** 7 mg/kg/dose PO q8h (max : 300 mg/dose) ± **Gentamicin** IV[4] for first 2 weeks	6 weeks	

1. Refer to Antimicrobial Dosing Recommendations for Pediatric Patients
2. If candidate for outpatient therapy, may consider administration via continuous infusion (same daily dose)
3. Refer to Empiric Dosing and Monitoring Recommendations for Vancomycin in Pediatric Patients
4. Refer to Empiric Dosing and Monitoring Recommendations for Aminoglycosides in Pediatric Patients

Last revised: 03/2021	**Antimicrobial Stewardship Subcommittee:** 11/2018	**P&T:** 11/2018
References		
	Citation	
Revision History		
Date	**Revision**	
03/2021	Updated vancomycin and gentamicin hyperlinks	

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.